

Representation-Level Uniqueness for Support-Constrained Cayley Graphs on the Symmetric Group

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1 The original conjecture

For integers $1 \leq r < k < n$, let

$$C(n, k; r) = \{\sigma \in S_n : \sigma \text{ is a } k\text{-cycle and } [r] \subseteq \text{supp}(\sigma)\}.$$

The set is closed under inverses, so it defines an undirected Cayley graph $\text{Cay}(S_n, C(n, k; r))$. The graph is usually nonnormal: the connection set is stable under permutations that preserve $[r]$, but generally not under all of S_n .

Irreducible representations of S_n are indexed by partitions $\lambda \vdash n$. Write ρ_λ for the corresponding representation and

$$T_\lambda(n, k, r) = \sum_{h \in C(n, k; r)} \rho_\lambda(h).$$

In an orthogonal realization this is a real symmetric operator. The adjacency spectrum of the whole Cayley graph is obtained by taking $\dim \rho_\lambda$ copies of the spectrum of every block $T_\lambda(n, k, r)$.

The standard representation is indexed by $(n-1, 1)$. Siemons–Zalesski and Li–Xia–Zhou computed its largest eigenvalue as

$$\mu_2(n, k; r) = \frac{(k-2)!}{n-r} \binom{n-r}{k-r} \left((k-1)(n-k) - \frac{(k-r-1)(k-r)}{n-r-1} \right). \quad (1)$$

They also proved that this is the relevant second eigenvalue for the range used below. Their sharper question was which irreducible blocks attain it.

Conjecture 1 (Li–Xia–Zhou, Conjecture 4.7). *Suppose $n \geq 5$ and $1 \leq r < k \leq n-2$.*

1. *If k is even, the second largest eigenvalue is attained only by the standard representation $(n-1, 1)$.*
2. *If k is odd, the strictly second largest eigenvalue is attained exactly by $(n-1, 1)$ and its sign twist $(2, 1^{n-2})$.*

The argument below addresses this uniqueness statement throughout its stated range. The central idea is to make the endpoint of the published induction strict and then carry that strictness through the same recurrence. Two concrete models rule out equality at the endpoint; small cycle lengths and three finite boundary cases are handled separately.

The word “strictly” matters in the second clause. An odd k -cycle is an even permutation, so the graph has two components and the degree occurs twice. When k is even the graph is connected. More precisely, $C(n, k; r)$ generates S_n for even k and A_n for odd k . Thus the degree block is just the trivial representation in the first case, and consists of the trivial and sign representations in the second.

Theorem 2 (Complete uniqueness theorem). *The conclusion of Conjecture 1 holds for every parameter triple in its stated range.*

2 What the proof needs from earlier work

The proof separates a known inequality from the new equality analysis. For $4 \leq k \leq n - 2$, Li–Xia–Zhou proved that the largest eigenvalue of every nondegree irreducible block is at most $\mu_2(n, k; r)$. Their argument did not say when equality could occur. We will refer to this as the *published non-strict bound*.

We also use three boundary results from the same paper.

1. For the layer $k = m - 1$ with m odd, the required uniqueness is known when $m/2 < r < m - 1$.
2. For $k = m - 1$ with $m \geq 6$ even, the non-strict bound is known for all $(m, r) \neq (6, 1)$, and uniqueness is known for $3 \leq r \leq m - 2$.
3. For even $n \geq 8$, uniqueness in the layer $k = n - 2$ is known when $1 \leq r < (n - 1)/2$.

These are inputs, not consequences of the argument below.

Three elementary facts organize the rest of the proof.

Lemma 3 (Equality in Weyl’s inequality). *If A and B are real symmetric operators, then*

$$\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B).$$

Equality holds precisely when A and B have a common nonzero top eigenvector.

Proof. Take a unit top eigenvector v of $A + B$. Its Rayleigh quotient is the sum of the two Rayleigh quotients for A and B , each bounded by the corresponding top eigenvalue. Equality in the sum forces equality in both terms. The converse is immediate. $\square$

The restriction of a Specht module from S_n to a point stabilizer S_{n-1} is multiplicity free: its children are obtained by deleting one removable box. In particular, the standard child $(n - 2, 1)$ can occur only under

$$(n - 1, 1), \quad (n - 2, 2), \quad (n - 2, 1, 1). \tag{2}$$

The analogous statement for the sign-twisted standard child is obtained by conjugating these partitions.

Finally, let Y be an m -element set and let K_Y be the complete class of m -cycles supported on Y . It has $(m - 1)!$ elements. In any orthogonal representation,

$$\lambda_{\max} \left(\sum_{g \in K_Y} \rho(g) \right) \leq (m - 1)!. \tag{3}$$

Equality holds exactly on vectors fixed by S_Y when m is even and by A_Y when m is odd. Indeed, every summand has Rayleigh quotient at most one, and equality in their sum fixes the vector under every m -cycle; those cycles generate the stated group.

3 How strictness propagates

Choose $t = r + 1$. A cycle in $C(n, k; r)$ either fixes t , or moves it. When $r \leq k - 2$, this gives the disjoint split

$$C(n, k; r) = C(n - 1, k; r) \dot{\cup} C(n, k; r + 1), \quad (4)$$

after relabelling the points other than t in the first part. On an irreducible parent module, the first operator is block diagonal over the children under restriction to S_{n-1} .

Lemma 4 (Strict propagation). *Assume uniqueness is already known for $(n, k, r + 1)$, and the published non-strict bound holds for $(n - 1, k, r)$. Then uniqueness holds for (n, k, r) .*

Proof. Consider a parent block that is neither a degree block nor one of the target blocks. It has no trivial child; for odd k it also has no sign child. The first summand in (4) therefore has top eigenvalue at most $\mu_2(n - 1, k; r)$. The induction hypothesis makes the second summand strictly smaller than $\mu_2(n, k; r + 1)$. Weyl's inequality is consequently strict. The target values satisfy the published recurrence

$$\mu_2(n, k; r) = \mu_2(n - 1, k; r) + \mu_2(n, k; r + 1),$$

so the parent block is strictly below $\mu_2(n, k; r)$. $\square$

Thus all that is missing is a strict starting point at $r = k - 1$.

4 The strict endpoint $r = k - 1$

At the endpoint, take $t = k$ and $Y = [k]$. The split is now

$$C(n, k; k - 1) = C(n - 1, k; k - 1) \dot{\cup} K_Y, \quad (5)$$

where K_Y is the full class of k -cycles on Y . Formula (1) becomes

$$\mu_2(n, k; k - 1) = (k - 1)!(n - k). \quad (6)$$

The possible equality vectors in (5) are ruled out by two concrete models.

Lemma 5 (The two-subset model). *Let $N \geq 6$, let $t \in [N]$, and let $Y \subset [N]$ contain t , with $4 \leq |Y| \leq N - 1$. Inside $S^{(N-2,2)}$, the standard child for the point stabilizer of t contains no nonzero A_Y -fixed vector.*

Proof. Realize $S^{(N-2,2)}$ inside the permutation module on two-subsets as symmetric arrays $x_{ij} = x_{ji}$, with zero diagonal and zero row sums. The standard child consists of arrays determined by numbers u_i , $i \neq t$, with $\sum_{i \neq t} u_i = 0$:

$$x_{ij} = u_i + u_j \quad (i, j \neq t, i \neq j), \quad x_{it} = -(N - 3)u_i. \quad (7)$$

First, A_Y -fixed and S_Y -fixed vectors coincide in the ambient two-subset module. On restriction to S_Y , this module is a sum of the permutation modules on two-subsets of Y , on pairs with one point in Y , and on pairs outside Y . Their stabilizers contain odd permutations, so none contains the sign representation. Since the A_Y -fixed space is the sum of the trivial and sign isotypic parts, only the trivial part remains.

Now suppose the array (7) is S_Y -fixed. The entries on two-subsets of Y are all equal. Comparing an entry x_{ab} , with distinct $a, b \in Y \setminus \{t\}$, with x_{at} shows that all u_a have a common value c and $2c = -(N - 3)c$. Hence $c = 0$. For $o \notin Y$, invariance under a permutation exchanging a and t gives $u_o = -(N - 3)u_o$, so $u_o = 0$. Thus every u_i vanishes. $\square$

Lemma 6 (The exterior-square model). *Under the same hypotheses, the standard child inside $S^{(N-2,1,1)}$ contains no nonzero A_Y -fixed vector.*

Proof. Let

$$V_N = \{(z_1, \dots, z_N) \in \mathbb{C}^N : \sum_i z_i = 0\}$$

be the standard module. The module $S^{(N-2,1,1)}$ is $\bigwedge^2 V_N$. Put

$$w = (N-1)e_t - \sum_{i \neq t} e_i.$$

Under the point stabilizer of t , the standard child is precisely the space of $u \wedge w$, where $u_t = 0$ and $\sum_{i \neq t} u_i = 0$. Its antisymmetric coefficients are

$$c_{ij} = u_j - u_i \quad (i, j \neq t), \quad c_{it} = (N-1)u_i. \quad (8)$$

Again the A_Y - and S_Y -fixed spaces coincide. Indeed, on restricting to S_Y , the exterior square is a sum of $\bigwedge^2 V_Y$, copies of V_Y , and trivial modules; none is the sign module because $|Y| \geq 4$. If $u \wedge w$ is S_Y -fixed, the transposition $(a \ t)$, $a \in Y \setminus \{t\}$, reverses the coefficient on $\{a, t\}$. Hence $c_{at} = 0$ and $u_a = 0$. For $o \notin Y$, the same transposition compares c_{ao} and c_{to} and forces $u_o = 0$. Thus $u = 0$. $\square$

Proposition 7 (Strict endpoint). *For every $k \geq 4$ and $n \geq k+1$, the uniqueness conclusion holds at $r = k-1$.*

Proof. Induct on n . When $n = k+1$, the assertion is the published $k = n-1$ boundary result at $r = n-2$, in precisely a range where that result is strict.

For the induction step, use (5). On any parent that is neither a degree block nor a target block, the first summand is bounded by $\mu_2(n-1, k; k-1)$ by the induction hypothesis. Equality can occur only on a standard child, and, for odd k , possibly its sign twist. The second summand is bounded by $(k-1)!$ by (3); equality requires an S_Y -fixed vector for even k , or an A_Y -fixed vector for odd k .

Weyl's inequality and (6) give the desired non-strict bound. If equality held, the same vector would have to satisfy both equality conditions. Branching leaves only the two nontarget parents $(n-2, 2)$ and $(n-2, 1, 1)$, together with their conjugates in the odd case. Lemmas 5 and 6 rule out the two parents. When k is odd, all generators are even, so sign twisting leaves the operator unchanged and rules out the conjugate parents as well. The inequality is therefore strict. $\square$

5 Cycle lengths two and three

The general published non-strict theorem starts at $k = 4$, so the two smaller lengths are checked directly. Let

$$X_m = \sum_{i < m} (i \ m)$$

be the m -th Jucys–Murphy element. In Young's orthogonal basis it acts diagonally: its eigenvalue is the content of the box containing m .

Proposition 8. *For every $n \geq 5$, the uniqueness conclusion holds in each of the cases*

$$(k, r) = (2, 1), \quad (3, 1), \quad (3, 2).$$

Proof. For $k = 2, r = 1$, conjugation identifies the connection sum with X_n . The standard block has top content $n - 2$. Outside the trivial and standard representations every removable corner has content at most $n - 3$, giving strictness.

For $k = 3, r = 1$, if $X = \sum_{a \neq 1} (1 \ a)$, then the products $(1 \ a)(1 \ b)$, $a \neq b$, run once through the three-cycles moving 1. Thus the connection sum is $X^2 - (n - 1)$. The target value is $(n - 2)^2 - (n - 1)$; outside the degree and target blocks, all relevant contents have absolute value at most $n - 3$, so the inequality is strict.

For $k = 3, r = 2$, conjugate the last two distinguished points to $n - 1, n$ and write $s = (n - 1 \ n)$. The connection sum becomes

$$s(X_{n-1} + X_n) - 1.$$

On each one- or two-dimensional tableau space paired by s , the sum $X_{n-1} + X_n$ is a scalar, namely the sum σ_T of the two box contents. The possible eigenvalues are $\sigma_T - 1$, $-\sigma_T - 1$, or the corresponding two signs in a two-dimensional pair. If the two boxes are the last two boxes of row i , the value is $2(\lambda_i - i) - 2 \leq 2\lambda_1 - 4$; the column case similarly gives $2(\lambda'_j - j) - 2 \leq 2\ell(\lambda) - 4$. Outside the degree and target representations these bounds are at most $2n - 8$. If the boxes lie in different rows and columns, their contents give $|\sigma_T| \leq n - 3$, because $\sigma_T \leq \lambda_1 + \lambda_2 - 3 \leq n - 3$ and the conjugate estimate gives the opposite inequality. Hence that eigenvalue is at most $n - 4$. Both bounds are strictly smaller than the standard value $2n - 6$. $\square$

6 Completion of the proof

Proof of Theorem 2. Proposition 8 handles $k = 2, 3$. Assume now $k \geq 4$.

If $n - k \geq 3$, Proposition 7 gives strictness at $r = k - 1$. Descend through $r = k - 2, k - 3, \dots, 1$ using Lemma 4. At every step, the first summand lies in the range $4 \leq k \leq (n - 1) - 2$, where the published non-strict bound applies.

It remains to treat $k = n - 2$. The endpoint $r = n - 3$ is again Proposition 7.

- If $n \geq 9$ is odd, the needed non-strict estimates for the $k = (n - 1) - 1$ child layer hold for every r ; strict propagation reaches all the way to $r = 1$.
- If $n = 7$, propagation reaches $r = 2$. The remaining case $(7, 5, 1)$ is one of the exact computations in Appendix A.
- If $n \geq 8$ is even, the published $k = n - 1$ boundary result allows propagation down to $r = n/2$. The remaining values $1 \leq r \leq n/2 - 1$ are exactly the already proved even- n , $k = n - 2$ range listed in Section 2.
- If $n = 6$, the endpoint is $r = 3$, while the two remaining cases $(6, 4, 1)$ and $(6, 4, 2)$ are the other exact computations in the appendix.

These cases exhaust $n \geq 5$ and $1 \leq r < k \leq n - 2$. $\square$

7 The codimension-four case

The earlier public note proved only the slice $n - k = 3$. The strict endpoint argument above no longer depends on the codimension, so it proves every layer simultaneously once the published non-strict estimates and boundary cases are inserted. In particular it gives the following specialization.

Corollary 9 (The $n - k = 4$ slice). *Let $n \geq 6$, put $k = n - 4$, and let $1 \leq r < k$. If n is even, only the standard representation attains the second eigenvalue. If n is odd, exactly the standard representation and its sign twist attain the strictly second eigenvalue.*

Proof. Here $k = n - 4 \leq n - 2$, and k has the same parity as n . Apply Theorem 2. $\square$

The full theorem also covers every other legal codimension. It does not address the distinct $k = n - 1$ problems in the source paper.

A The finite boundary

The proof uses exact computation only for

$$(n, k, r) = (6, 4, 1), \quad (6, 4, 2), \quad (7, 5, 1).$$

The companion verification package in [5] supplies a self-contained implementation. It builds every irreducible representation in Young’s seminormal form over $\mathbb{Q}$, checks the Coxeter relations and transposition characters, sums the connection set exactly, and computes each characteristic polynomial over $\mathbb{Q}[x]$. After extracting the factor $(x - \mu_2)$ with its exact multiplicity, it uses Sturm root counts to exclude roots strictly above μ_2 . The results are:

(n, k, r)	degree	μ_2	attaining blocks	block multiplicity
$(6, 4, 1)$	60	18	$(5, 1)$	4
$(6, 4, 2)$	36	16	$(5, 1)$	3
$(7, 5, 1)$	360	84	$(6, 1), (2, 1^5)$	5

Every block other than the degree and target blocks has no root at or above μ_2 . The target blocks have the displayed multiplicities and no root above μ_2 .

For comparison, the original proof draft reports additional regression data for $(6, 4, 3)$ and $(7, 5, r)$, $2 \leq r \leq 4$:

(n, k, r)	degree	μ_2	attaining blocks	block multiplicity
$(6, 4, 3)$	18	12	$(5, 1)$	2
$(7, 5, 2)$	240	78	$(6, 1), (2, 1^5)$	4
$(7, 5, 3)$	144	66	$(6, 1), (2, 1^5)$	3
$(7, 5, 4)$	72	48	$(6, 1), (2, 1^5)$	2

These four rows are historical regression data, not proof obligations; they are not part of the new three-case verification.

The companion `extension/check.py` regenerates `extension/results.json`, which records every block’s characteristic polynomial, threshold multiplicity and root count. An independently constructed orthogonal model also checks the spectra numerically; the finite proof inputs rely on the rational calculations, not on floating-point agreement. These computations certify only the three stated boundary cases. The general argument remains the proof given in the preceding sections, using the published results cited there.

References

- [1] L. Flatto, A. M. Odlyzko and D. B. Wales, *Random shuffles and group representations*, Annals of Probability 13 (1985), 154–178.
- [2] X. Huang and Q. Huang, *The second largest eigenvalues of some Cayley graphs on alternating groups*, Journal of Algebraic Combinatorics 50 (2019), 99–111.
- [3] Y. Li, B. Xia and S. Zhou, *The second largest eigenvalue of some nonnormal Cayley graphs on symmetric groups*, Journal of Combinatorial Theory, Series A 218 (2026), 106097, [doi link](#).
- [4] J. Siemons and A. Zalesski, *On the second largest eigenvalue of some Cayley graphs of the symmetric group*, Journal of Algebraic Combinatorics 55 (2022), 989–1005.
- [5] EULER public proof and companion verification package, [PR 17](#), original proof draft at commit `5a2457a461196e806e07222d57b6290b7d5aeb04`; the accompanying verification files are supplied with the updated pull request.